

AGRI-PV NAVIGATOR

Comprehensive Techno-Economic & Agrivoltaic Feasibility Report

Project Proposal: Ganga Valley Farm (Updated)

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Report Type: PROPOSAL | Verification Standard: CERC / MNRE Compliant

1. FARM & GEOGRAPHICAL PROFILE

Farm Name:	Ganga Valley Farm (Updated)	Total Land Area:	3.00 Acres
Location:	Phulpur, Prayagraj, Uttar Pradesh, India	Primary Crop:	Wheat
Soil Type:	Loamy	Topographical Slope:	1.8% (Almost flat)
Irrigation Source:	Available	Grid Distance:	2.2 km to Feeder

2. SITE SUITABILITY & SOLAR POTENTIAL

Overall Suitability Score: 86/100 (Highly Suitable)

Factor	Score	Metric Value	Impact Evaluation
Solar Resource	92/100	4.8 kWh/m²/day	Positive: Projected high generation and sh
Land Slope	90/100	1.8% slope	Positive: Low civil and site preparation e
Soil Type	85/100	Loamy soil	Positive: Strong foundation anchoring and
Water Availability	80/100	Irrigation available	Positive: Regular soiling prevention.
Crop Shade Tolerance	82/100	Wheat (C3 crop)	Positive: Microclimate humidity reduces wa
Grid Proximity	85/100	2.2 km to feeder	Positive: Low electrical line expansion co

3. AGRI-PV SYSTEM CONFIGURATION

Installed DC Capacity:	275.0 kWp	Annual Generation:	445.4 MWh/yr
Mounting Structure:	ELEVATED	Panel Tilt Angle:	20° South-Facing
Module Clearance Height:	2.8 m	Inter-row Spacing:	6.0 m
Land Coverage Ratio:	40%	Cultivable Land Retained:	81.3%

4. FINANCIAL & ENVIRONMENTAL PROJECTIONS (25-YEAR LIFETIME)

Estimated Project Capex:	Rs. 1.40 Crore	Annual Combined Revenue:	Rs. 12.5 Lakh/yr
Simple Payback Period:	10.0 Years	25-Yr Net Present Value (NPV):	Rs. 10.0 Lakh
Land Equivalent Ratio (LER):	1.59x (Productivity Boost)	Annual CO2 Emissions Offset:	365 Metric Tons/yr
Internal Rate of Return (IRR):	16.4%	Machinery Clearance:	Tractor (2.8m) Harvester requires bypass